

06-15-25

CSC 659 859 Summer 2025

Team Project overview and instructions

**Develop ML or GenAI application
then audit it for AI Ethics and
Trustworthiness
35/100 points**

Prof. D. Petkovic

High level goals

- Creating Ethical and Trustworthy AI systems that do good for humanity is one of the most important things we can do
 - this team project helps you learn how to develop and audit AI application and audit for trustworthiness and ethics
- **Goals: perform Ethics and Trustworthiness audit of either a classical ML system or GenAI system**
 - You will NOT be summarizing work of others but will do your own development and audit
- Work is done by your student team – helps learn how to work in a team of people with different backgrounds → critical skill today e.g. for adoption of new AI technologies
- Great experience and portfolio for job search
- **Work to be started after teams are formed**
- Instructor will help with advice, and organize steps and phases of the project

Objectives and goals of team project

- Apply material learned in CSC 859 in the context of a real life scenario/case study:
 - A team of 2-3 students represents a team made of **company R&D** (e.g. CS majors) and **domain/business experts** (e.g. non-CS majors) creating and then self-auditing a **chosen ML or GenAI application**
 - Team goal is to write and present a report to be used to convince company CEO (e.g. Prof. Petkovic) of viable ML/GenAI solution to a chosen problem and attest that the solution is ethical and trustworthy e.g. accurate, ethical, explainable, transparent, non-biased etc.
- Each student team creates one report
- Each team appoints team lead to organize work and communicate with instructor (e.g. sending reports)
- Each team presents their project to the whole class - 20 min/team - last class (and submits a final report for grading)
- **Grading: 35/100 points** – all team members get same grade points unless there are some issues (e.g. low participation)

Team Project: Classic ML or GenAI

- **Option A: develop and audit chosen classical ML application for ethics and trustworthiness**
 - Chose the data and application from open source databases
 - Develop RF Random Forest to chosen data/application
 - Audit developed application for ethics and trustworthiness
- **Option B: Develop and audit chosen GenAI/ChatGPT application for ethics and trustworthiness**
 - Chose application area (e.g. SW engineering, ML development, cooking, math, chatbot for some domain....)
 - Use free GenAI/ChatGPT or like as available (e.g. you do NOT chose training data)
 - Create test and evaluation database
 - (ChatGPT/GenAI fine-tuning is optional (if students have access))
 - Test GenAI application then audit it for ethics and trustworthiness
- **Write and present a report in class (one report per team)**
- **Teams can chose options A or B and the application/data but instructor has to approve before starting**

What do you gain from team project?

- Practical implementation of class material – ***development and audit of AI application***
- Learn to work and communicate with “others” NOT in your area
- Learn basic teamwork skills
- CS majors learn to work with non-ML experts who are often key decision makers and how to “convince them and increase their trust”
- Non-CS majors learn how to audit ML products
- Writing about and presenting professional work
- Obtain portfolio for job search (report, github) and improve your CV

About teams and project organization

- Teams are of generally 2-3 students
- Teams are selected by instructor based on skills survey in order to have class teams of appr. similar expertise profile
 - Team will be preferably of students with mixed level of skills and mixed majors (e.g. CS and non-CS majors) if possible – great for student experience and simulates real world
- Team appoints team lead who communicates with instructor, organizes meetings, ensures work gets done, checks and sends reports etc. This is in addition to regular work same as what all team members do
 - Team lead gets up to 3 class points for job well done
- Teams work last 30-45 min in Wednesday class. Also strongly recommended for team to find one more slot to meet outside of class

Teamwork responsibilities

- ALL team members must attend mandatory team sessions or send e-mail with valid excuse reason (to team and instructor)
- ALL team members must contribute to the best of their abilities
- Team lead provides project management and follow up and must ensure that all team members get concrete tasks and deliver on time
- Team lead must report any lack of attendance or engagement in their team to the instructor asap
- If student can not attend must send e-mail to team and instructor with reasons for excuse
- **Poor attendance and engagement will result in reduced team grade for that individual**

Team report outline

- Team works together on ONE team report
- Work presented at the end of the class
- We provide outlines for the reports: **Option A** – classical ML audit; **Option B** – GenAI application audit
- Report outline **MUST** be strictly followed
 - Section titles are in **bold**
- Outline is modeled by professional reports and technical papers

Student team projects – outline of final report –

Option A – classic ML

Title page (title, team member names with e-mails, date)

1. **Executive summary** (1 page max)
2. **Motivation, problem description and case study goals**
3. **Data:** Chose, describe and audit the data to be used – description, analysis, full database audit and analysis etc. – use class slides as a guidance
4. **ML Methods:** Describe RF methods to be used chosen data using SciKit of R - follow ML best practices as in HW 2
 - Tools and methods used
 - Parameters for setup, training, accuracy evaluation , explainability analysis etc.
5. **ML Accuracy Results:** Present full accuracy and explainability results (similar to HW 2). Adhere to full provenance
6. **Audit for ethics and trustworthiness** (describe methods /measures and then implement them in your audit, cover each audit component)
7. **Summary and recommendations** (understandable to ML and non-ML but domain experts)
8. **App. I: Summary of each team member contributions** (max1/2 page per member)
9. **App. II: Code used for ML experiments** (in PDF format) issues)
10. **App. III: Describe and self-asses related GenAI usage for getting help on the project**
11. **References**

Student team projects – outline of final report –

Option B – GenAI

Title page (title, team member names with e-mails, date)

1. **Executive summary** (1 page max)
2. **Motivation, problem description and case study goals**
3. **GenAI technology and application to be evaluated** - for what application GenAI will be evaluated, describe the data it was trained on (from the open literature)
4. **Test and verification data:** describe details and how you developed this
5. **Methods of GenAI accuracy evaluation** to be used (use material from the class on AI e.g. develop test questions and rank GenAI responses and analyze results). If fine-tuning used describe it separately
6. **Results of accuracy evaluation experiments** (e.g. show all test questions and responses) and summarize results (e.g. statistics of evaluation of GenAI responses wrt. to your test questions)
7. **Audit for ethics and trustworthiness** (describe methods /measures and then implement the audit, cover each audit component)
8. **Summary and recommendations** (understandable to ML and non-ML but domain experts)
9. **App. I: Summary of each team member contributions** (max1/2 page per member)
10. **App. II: Tools and code used for ML experiments** (in PDF format)
11. **App. III: Describe and self-asses related GenAI usage for getting help on the project**
12. **References**

Title page

**CSC 659 859 AI Ethics and Explainability
San Francisco State University**

Summer 2025

**Team project: Development and Ethics and Trustworthiness Audit of AI
Application**

Team N

<member names> <sfsu e-mail: (team lead first)

Date

More details for Option A team project: classic ML

Section 1 (Options A and B): Executive summary - max one page

- What is the goal of this report (one paragraph)
- Problem description and its importance (one paragraph)
- Data available and technologies and tools used (one paragraph)
- Methods of audit (one paragraph)
- Summary of results
- Recommendations of your work (one paragraph) – system is Ethical and Trustworthy? OK to use?
 - (NOTE: it is OK to NOT recommend the system but have to justify it)

Section 2 for Option A – classical ML: Motivation, problem description and case study (appr. 2 pages)

- Choose problem with good and adequate data available
- Go to open source data set repositories like UC Irvine – describe the data and related application
 - <https://archive.ics.uci.edu/>
- Motivation: why the chosen problem is important, why needs ML and audit. Describe the importance, impact to society, data you will use, what decisions ML has to make and are they risky/impactful
- Describe specific goals of your study

Section 3 for Option A: Data and data audit (2-3 pages)

- **Note: Most biases in ML come from data. Analyzing data is critical part of AI audit**
- Choose problem with good and adequate data available
- Go to open source data set repositories like UC Irvine
 - <https://archive.ics.uci.edu/>
- Describe the data in sufficient detail
- Perform FULL Audit of training data before deciding (consult class slides – address all steps of data audit), e.g.:
 - Is the data format/type applicable for ML you will use (note: RF uses tabular data and does not work well for signals and images)
 - Data well explained?
 - Minimal or no missing data?
 - Enough data to cover all demographics?
 - Ground truth established (necessary for supervised ML)?
 - Goals of ML decisions? How risky, impactful?

Section 4 for Option A: ML Methods (size as needed, make sure you provide necessary details)

- Describe methods and tools to be used, including ChatGPT
- How is the data organized
- What ML and tools will be used (use RF, with R or SciKit)
- How will RF be trained: describe methods and all hyperparameters values for e.g. grid search.
- How will accuracy be evaluated:
 - What accuracy measures will be computed (do ALL)
 - What accuracy estimation method will be used: OOB or CV or both
- How will explainability be evaluated?
 - Use some measures from Prof. Petkovic RFEX paper or LIME
 - How will you relate explainability with domain ground truth
- How will ChatGPT be used (if any)

(Here you tell us what you WILL be doing, do not present actual results here)

Section 5 for Option A: ML Accuracy Results

(size as needed, provide all necessary details)

- Present full accuracy and explainability results (as in HW 2)
- Present result for ALL accuracy measures
- Provide FULL description of parameters and options use for EACH result/chart/graphics – full provenance needed
- Use what you learned in HW 2 to make report easy to read, well organized and provide full data provenance (this is also graded)

Section 6 Option A : Audit for ethics and trustworthiness (size as needed, provide all necessary details)

- Describe AI Ethics and Trustworthiness Methods to be used in this report – chose one method from the class and summarize it and give reference to it
 - Aim to cover main components like accuracy and risk/impact of errors, bias, fairness, risk, human control, privacy, transparency/explainability
- Apply audit methods the best you can
- Describe WHAT you have done and show results for each Ethics and Trustworthiness comments separately
(This is not easy, it is still subjective, so do your best)

Options A and B: Summary and recommendations (up to 2 pages)

- Written for executives, domain experts and not just for ML experts
- Outline
 - Problem you were solving,
 - Data available, ML methods applied, technologies used
 - Summarize accuracy and explainability results
 - Summarize AI Audit results
 - Recommend whether this should be trusted/used or not and what are the major issues/risks or unsolved remaining issues
- NOTE: It is OK NOT to recommend the system, but have to explain why

Options A and B: Appendix I: Summary of each team member contributions

- Each team member sends to ALL team members e-mail with up to half a page of their contributions to team project (list/itemized format OK).
- Document editor puts emails of all team members together and puts in this section (MUST preserve full e-mail text including sender and receivers)
- Used for instructor to ensure each member contributed and the work was as evenly spread as reasonably as possible
- In addition for Team lead: in about one page describe challenges (technical, organizational, teamwork), how you addressed them, and what would you do better next time

Option A and B: Appendix II: Code used

- Submit main code snippets as PDF due to security reasons
- If Github used show its home page and describe its organization

For Option A and B: Appendix III: ChatGPT and related GenAI usage (up to 2 pages)

If ChatGPT or related tool was used please describe:

- What version you used
- List for all specific tasks you used it
- For each listed task say:
 - How it helped and how did you use it (max ½ page text but also desirable to use examples like tool output – text, code)
 - Rank how much the tools helped (LOW, MEDIUM, HIGH)
 - List main prompts you used
- Explain how you verified tool recommendations/output (remember - YOU MUST verify and test it, it is YOUR responsibility to be accurate)
- Reflect on how helpful the tool was overall

Options A and B: References

- Use standard way as in research papers – numbered list of resources used, referenced by the number in the text. For URLs provide URL, title and date accessed
- List GenAI and tools used – provide URL, version and date URL accessed
- Mention class slides by slides title
- List names of people outside of team who helped you

More details for Option B team project – Use of GenAI

Section 1 (Options A and B): Executive summary - max one page

- What is the goal of this report (one paragraph)
- Problem description and its importance (one paragraph)
- Data available and technologies and tools used (one paragraph)
- Methods of audit (one paragraph)
- Summary of results
- Recommendations of your work (one paragraph) – system is Ethical and Trustworthy? OK to use?
 - (NOTE: it is OK to NOT recommend the system but have to justify it)

Section 2 for Option B – GenAI : Motivation, problem description and case study goals - (appr. 2 pages)

- Describe chosen GenAI technology to be evaluated
- Choose application area you will evaluate (e.g. domain of GenAI application)
- Motivation: why the chosen problem is important, why needs ML and audit. Describe the importance, impact to society, data you will use, what decisions ML has to make and are they risky/impactful
- Describe specific goals of your study

Section 3 for Option B: GenAI technology and application to be evaluated (up to 3 pages)

- Describe for what application GenAI will be evaluated
- Describe specific GenAI technology and version
- Describe any customization to be done for the chosen application domain

Section 4 for Option B: Test and verification data

- Describe test and verification data
- These can be data or questions/prompts and ground truth responses to prompts

Section 5 for Option B: Methods of GenAI accuracy evaluation (size as needed, provide all necessary details)

- Describe methods and tools to be used to evaluate accuracy of GenAI for chosen application
- Consult class material on Auditor for GenAI. Ideas are below:
 - You could as one option develop set of text questions (50 minimum) – say how will you do it, how many, what is the design objective
 - Develop evaluation methods for GenAI response to test question e.g. ranking by team members
 - Develop methods to compare team ranking vs. GenAI response to establish some measure of accuracy
 - (Do all this in the context of your chosen application)

(Here you tell us what you WILL be doing, do not present actual results here)

Section 6 for Option B: Results of accuracy evaluation experiments (size as needed, provide all necessary details)

- Present all test results e.g. test questions used and ranking results for each, in an easy to read way (as a table?) accuracy and explainability results (as in HW 2)
- Present details of accuracy evaluation measures to compare GenAI for team member question ranking
- Provide FULL description of parameters and options used for EACH result/chart/graphics
- If you can come up with some explainability analysis describe it
- Use what you learned in HW 2 to make report easy to read, well organized and provide full data provenance

Section 7 Option B: Audit for ethics and trustworthiness (size as needed, provide all necessary details)

- Describe AI Ethics and Trustworthiness Methods to be used in this report – chose one method from the class and summarize it and give reference to it
 - Aim to cover main components like accuracy and risk/impact of errors, bias, fairness, risk, human control, privacy, transparency/explainability
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- List names of people outside of team who helped you

Tools for team project

- Teams decide what is best and set up their IT infrastructure. **Use only free tools or tools you already haven (e.g. school provided GenAI tools)**
 - (ChatGPT fine-tuning access and use optional)
- For Option A ML development SciKit or R are recommended, with Jupyter and R Markdown for export of data
- Final project is in one PDF file. Team project editing can be done with google docs or like, then export to PDF
- Use of GenAI for Option A is encouraged but must be documented – follow class policies
- Recommended to use Github to manage project documentation and code
 - Organize it well, this is your portfolio too!
 - Have informative read me, folder for project report, folder for data, folder for ML experiments etc.
- Helpful resources on SW tools posted on Canvas plus use class slides for guidance how to things
- Team internal comm. tools are up to the team, instructor is not involved

Team project report format

- One PDF file per team, professionally organized (title page and separate sections titled exactly as in outline slides in **bold**, and in that order)
- Report size: about 25 – 40 pages excluding Appendices, single space, font 12ppt
- OK to use specialized tools to show or export technical results (charts, tables, graphs) into report PDF like <https://rmarkdown.rstudio.com/> or <https://jupyter.org/>
- OK to use ChatGPT or related tools but must follow class policies and document it
- Appoint document editor to make sure document is a viable quality technical report and readable to no-AI experts – this is also part of grading

Code and document management

- Teams decide what is best and set up their IT infrastructure
- Use only free tools
- RF is efficient and can be run on a laptop, so no need for cluster HPC
- Recommended: Google docs for report writing, collaboration and editing
- Code (main parts of it): as PDF in Appendix is preferred (so it stays as documentation)
- Each main code segment MUST have a header: purpose, name of developer, date. Document key code blocks with 1-2 lines of comments. Document code shown in Appendix
- Recommend use of Github as a general project repository (report and code— good as your e-portfolio) PLEASE do not post class slides in github

Team project grading – 35/100 points

- Adherence to project requirements and outline including section title and order – **2 points** (all sections must be submitted)
- Technical and scientific accuracy of ML and RF methods and actual experiments, same for GenAI application – **15 points**
- Quality of AI ethics evaluation and audit and adherence to related best practices – **15 points**
- Organization of the report including formatting and presentation (including how CatGPT or related tools were used) which must be at graduate level quality, and also easy to read and easy to understand by executives, domain experts and constituency who are not AI experts. Quality of full data provenance e.g. full descriptions for charts, tables, results. Proper use of references. Proper use of GenAI tools in Option A – **2 points**
- Quality of project class presentation: delivery, presentation content, time management, answering questions – **1 point**
- (Quality of general English is not graded as long as it is reasonable)
- **TOTAL 35/100 class points**

- **All team members get same grade unless there are exceptions like low participation (must be documented)**
- **Team lead may get up to 3 additional class points for job well done**

Roadmap

(N is your team number)

Phase I - Team Project Summary Plan – to be reviewed and approved by Prof. Petkovic

- Plans for team project MUST be approved by Prof. Petkovic before work continues – just to make sure you are on the right track (as done in real life)
1. Draft a report with Title page, Section 1 and Section 2 as in project outline. Put this in one PDF file
 - In Section 1 omit conclusions and recommendations
 2. Submit for approval is by Team lead: one PDF file per team sent by e-mail to Petkovic@sfsu.edu:
 - E-mail subject line “CSC 659 859 Summer 2025 team N – Team Project Summary Plan for approval”
 - Attached PDF file name “CSC 659 859 Summer 2025 team N – Team Project Summary Plan for approval”

Phase I - Team Project Summary Plan – approval and proceeding to work

1. Instructor will give feedback to Phase I proposed plan (e-mail and/or in person during class teamwork session)
2. Team revises sections 1 and 2 as per comments. Discuss with instructor as necessary (if everything is clear to you instructor does not need to review revised versions)
3. Use/build on above Sections 1 and 2 (after revision) for your final team project submission (ok to do more revs later like adding conclusions and recommendations in Section 1)
4. If everything is clear and approved proceed to the work
5. If questions ask early before investing in a lot of work (always a good – iterations with expert/management feedback)

Final report submission and presentation of team project for grading – last class (finals)

1. **Presentation to the class:** Team project findings to be presented to the whole class and Prof. Petkovic 20 min/team (can use PPT)
 - Each team present project in about 15 min, 5 min for Q&A. Can use PPT for presentation or go through the team report pieces
 - **Attendance to all pres. Is mandatory**, excuse has to be approved ahead of time
2. **Team report submission:** Team project report to be submitted by team lead via e-mail the day of project presentation to Petkovic@sfsu.edu :
 - E-mail subject line: “CSC 659 859 Summer 2025 - team N – team project final report”
 - Attached File: PDF with name “CSC 659 859 Summer 2025 - team N – team project final report”
 - N is your team number
 - Put some courtesy text in the body of e-mail
 - **Make copy of the report for yourself (including hard copy), this is your e-portfolio, helps in job search**
 - **Post your work in your github for your e-portfolio**

Enjoy!

- **Creating Ethical AI systems that do good for humanity is one of the most important things we can do**
- Learning benefits of team project:
 - Learn how to develop full AI application and audit it for trustworthiness and ethics
 - Learn how to work in a team of people and also in a team of people with different backgrounds
 - Learn how to *write and present* professional reports
 - **Create your e-portfolio for job search consisting of:**
 - a) team report; b) github with code and work examples; c) update CV with new skills set**